Master's Degree Program in

Data Science and Advanced Analytics

Business Intelligence

Retail4All

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Executive summary

Business Intelligence solutions are becoming an important competitive tool for organizations, as they enable companies to transform large volumes of data into meaningful insights, supporting more efficient data-driven decisions.

Retail4All is a retail company that operates through multiple stores and manages a large volume of daily transactional data. However, the company lacks a centralized system that aggregates different data sources and effectively analyses sales performance, making it difficult to obtain a clear view of the business.

Therefore, Retail4All aims to develop a BI system that allows for a multidimensional analysis of the business over time. To address this challenge, the Kimball methodology was used to develop a dimensional data model that integrates multiple data sources into a structured star schema. The model enables the analysis of key business metrics across different levels, improving performance monitoring, and enhancing operational efficiency.

The ETL process was then implemented to extract, transform and load data from multiple sources into a fully structured data warehouse, ensuring data quality and consistency throughout. The pipeline was designed to orchestrate the entire process in a controlled and efficient manner, maintaining referential integrity at all stages. Additional developments were incorporated to further enhance the solution, including an incremental load strategy, automated monitoring mechanisms and an advanced analytics component that enriches the data warehouse with new insights, strengthening Retail4All's analytical capabilities.

Finally, a reporting layer was developed on top of the data warehouse to turn this structured data into actionable insights. A semantic model was built with the appropriate hierarchies, data types, formatting and relationships, complemented by a set of DAX measures, including time-intelligence calculations, designed to answer the defined business questions. These measures feed an interactive Power BI report, organized into thematic pages, each with its own narrative, covering product, operator, channel and location performance. Together, these components give Retail4All's decision-makers an intuitive, self-service tool to monitor performance and support faster, more informed strategic and operational decisions.

Introduction

Retail4All is a prominent retail organization in Portugal that manages a vast network of physical and digital stores. Although the company processes a high volume of daily transactions and possesses three years of historical data, it currently lacks a centralized system to transform raw operational data into actionable insights.

To bridge this gap, this project implements a business intelligence solution using the Kimball methodology and a star schema architecture. This model enables the administration to perform advanced time-intelligence analysis, such as year-over-year sales comparisons and product and store performance tracking. Ultimately, this structured approach eliminates informational silos and empowers **Retail4All** with a faster, data-driven decision-making process.

Finally, we developed and implemented the ETL process, detailing the extraction, transformation, and loading of data from multiple sources into a structured data warehouse. It covers the pipeline architecture designed to orchestrate data flow, the transformation steps applied to ensure data quality and consistency, and the additional developments incorporated to enhance the overall solution. Together, these components lay the technical foundation for the analytical capabilities that will support Retail4All's strategic and operational decision-making.

Building on this technical foundation, this final stage focuses on the reporting layer of the solution. We developed a semantic model on top of the data warehouse, defining hierarchies, data types, formatting and the relationships needed to support the analysis, and created a set of DAX measures, including time-intelligence calculations, to answer the business questions defined at the start of the project. These measures feed an interactive Power BI report, organized into thematic pages, each with its own narrative, that together translate the company's data into clear, actionable insights. This reporting layer is where the value of the entire pipeline becomes tangible, giving Retail4All's decision-makers an intuitive, self-service tool to monitor performance across products, operators, channels and locations, and to support faster, data-driven decisions.

Business context

Retail4All is a dynamic company operating in the retail sector in Portugal. It is focused on providing a wide variety of consumer products to its customers. The organization conducts its daily business through an extensive network of physical and online stores.

To ensure a smooth operation, **Retail4All** relies on a diverse product catalog (organized by categories and subcategories**)**, a structured network of supply points (POS), and a dedicated team of operators who handle daily sales transactions. Furthermore, the company's activities are distributed across multiple geographic locations, generating a continuous and large volume of daily sales records.

The main objective of this project is to gather and organize this raw daily data, transforming it into a structured format that enables the company's management to clearly understand its overall performance, improve its analytical output and facilitate faster decision-making for business processes.

Business problem

Retail4All operates in a highly competitive market, where large volumes of data are extracted directly from the main operational servers of the company. The server is refreshed daily with new data, which allows us to perform updated analysis and faster decision-making. Although the company has three years of historical data, it lacks a centralized system to analyze and present key information such as sales trends over time, comparisons between years and performance by store or product category. This makes it hard to gain a clear and consistent view of business performance.

Additionally, the company is not fully able to analyze detailed transactions or build customer profiles, limiting its ability to understand clients and improve sales strategies. There is also a need to ensure data security by restricting access based on user roles and locations.

Therefore, the goal of the business intelligence solution is to design a data model that integrates the different data sources by filling the needs of the company.

Business Questions

To guide the development of the dimensional model and support the analytical solution, a set of clear and objective business questions was defined. These questions focus on understanding sales performance, operational efficiency, and the impact of different business dimensions such as products, stores, locations, and operators. In line with this, the business questions to be addressed include:

	Business Question	Measures
(1)	What are the total revenue and units sold by product category across different time periods (month, quarter, and year)?	Total revenue (sales) Units sold (quantity)
(2)	Which employees (operators) generate the highest sales across months, quarters, and years?	Sales
(3)	Which Points of Supply generate the majority of total sales?	Sales
(4)	How does sales volume differ between physical and online stores over time?	Sales
(5)	Which locations drive the highest revenue and growth?	Sales
(6)	How does store performance vary when accounting for store size, customer traffic, and competition level?	Sales

Table 1: Business questions.

Data Source

During the development of the project, initial data sources were provided, containing information on multiple dimensions of the business, such as sales, locations, products, points of supply, operators, and stores. To enhance the quality of the analysis, we felt the need to add some attributes to those tables to extract deeper business insights. This additional data was generated using artificial intelligence.

A summary description of the tables that will be used can be found below.

Sales (Retail)

The first table that was provided was the **sales** table, which centralizes information about every sale. This is the table with the largest number of records, as it details every sale made by Retail4All. The table has multiple attributes, including the **store ID**, **datetime**, indicating the date and time of each transaction, the **product identification**, the **store**, the **point of supply**, the **location**, the **currency** in which the payment was made, the **quantity** of products, the total **amount**, and the **ID of the operator** who processed the sale.

The Sales table reflects the core business of Retail4All, referencing information from multiple dimensions, it enables a complete analysis over time of the business performance and supports decision making at both operational and strategic level.

Locations

The l**ocation** table is used to record the geographical data for Retail4All sales points. The table indicates the **location id**, **city**, the **district**, and the **country**, which currently has one unique value ("Portugal"). From an initial overview, some missing values were identified in the district columns.

To allow for broader aggregation, a **region** attribute was introduced to categorize locations into North, Center, South and Islands. Simultaneously, a **store count per city** attribute was created, reflecting market density and store distribution. A **postal code** variable was also added, enabling geographic mapping.

The final location table will allow for regional performance analysis and comparison between different areas of the country, providing a solid foundation for decisions related to marketing and logistics management.

Products

The **products** table lists the products that are sold by **Retail4All**. In addition to the **product name** and its **unique business identifier**, the table contains details about each **product's category** and **subcategory**.

In terms of product analysis, creating a price dimension was significantly valuable to perform financial analysis, particularly regarding product profitability. Therefore, **sales price**, u**nit cost**, and the gross **margin** were added to the original table.

Stock information was also included to evaluate whether the product meets customer demand. The new attribute of **stock items**, representing the quantity of each product currently available, provided insights into inventory levels.

For categorical analysis, a **brand** attribute was added to understand customer preferences and brand performance. Additionally, a discontinued flag was included for identification of products that are no longer available, ensuring that the analysis focuses on active items.

Point of Supply

The **point of supply** table stores information about Retail4All supply points, particularly their **names** and **email addresses**. This table was interpreted as the external Retail4All supplier channels.

After analyzing the available data, it became clear that adding new features could enhance the business insights that can be extracted from these points. The first attribute to be created relates to the **number of years** that the point is **operating** with the company, reflecting the stability and longevity of its contracts. The second attribute is a binary variable, indicating **whether the point is active**, allowing only active channels to be considered when analyzing current business performance.

In addition, a **rating** (1-5) was incorporated to measure the quality of experience across these channels. This metric reflects the perceived quality and satisfaction of Retail4All with the suppliers, highlighting areas for improvement and identifying the best performance channels.

Operators

The **operators** table includes information about the employees who handle sales transactions at Retail4All. It contains basic details such as **operator ID**, **name**, **email**, **gender**, **role,** and the **team** each employee belongs to.

To make this data more useful from an analytical perspective, we added several attributes, with the help of Artificial Intelligence. These include the **experience level** (junior, mid, or senior), **performance tier** based on the sales each operator generates, and the **years in the company**. We also added more descriptive and analytical features, such as **average monthly sales**, overall **customer rating** (1-5), **age**, **nationality** and the city of the store, which is linked to the l**ocation** table.

Additionally, we included indicators like **sales trends**, allowing us to understand whether an operator's performance is improving or declining over time.

Overall, these enhancements allow us to go beyond simple operational data and better understand employee performance, behavior and their impact on sales outcomes.

Stores

Lastly, the **stores** table, originally divided into two datasets (stores A and stores B), contains information about the different retail locations where Retail4All operates. It includes attributes such as **store ID**, **store name**, **location**, and **whether the store operates online or physically**.

To enhance the analytical value of this table, we also added several features, using the help of Artificial Intelligence. These include a **store tier** (from 1 to 5) reflecting overall customer appreciation, and the **average monthly sales**. We also added a simplified **store size** category (small, medium, large) and the **number of employees** working at each store.

Furthermore, additional performance rates and contextual indicators were included, such as **sales growth**, which allows us to assess whether a store's performance is improving or declining over time. We also introduced the average **daily customer traffic**, **store age**, and an estimated nearby **competition Level**.

Overall, these enhancements provide a more comprehensive view of store characteristics and performance, supporting deeper and more meaningful business analysis.

Dimensional Model

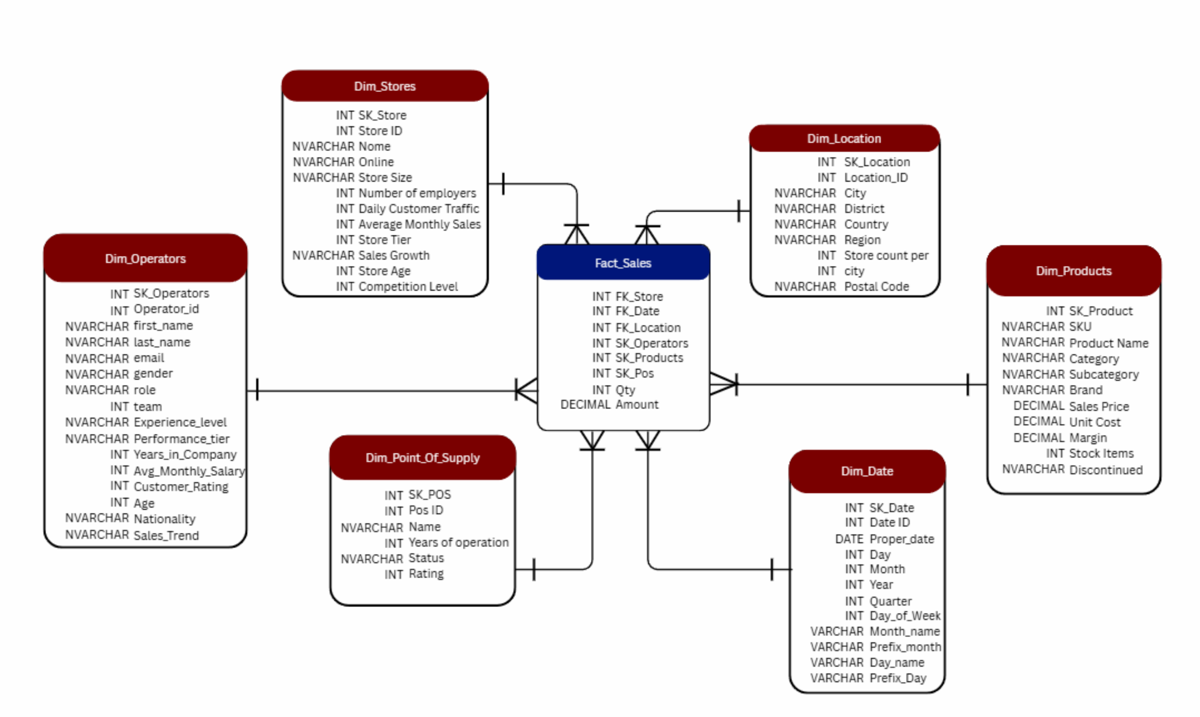

Figure 1: Star schema of **Retail4all**.

The dimensional model was designed using a star schema, with a central fact table (Fact_Sales) connected to several dimension tables. The **fact_sales** table contains the main transactional data that is linked to all dimensions through foreign keys, allowing analysis from different perspectives such as time, product, store, location, operator, and point of sale. The model includes the following dimensions: **Dim_Date**, which allows analysis over time (day, month, quarter, year); **Dim_Products**, which describes the products and their categories; **Dim_Stores**, which contains information about each store and its characteristics; **Dim_Location**, which supports geographic analysis (city, district, country, region); **Dim_Operators**, which provides information about employees and their performance; **Dim_Point_of_Supply**, which describes the different points of sale.

To support data analysis at different levels of detail, the following hierarchies were defined:

Time: Year → Quarter → Month → Day;

Product: Category → Subcategory → Product;

Location: Region → District → City.

Furthermore, as illustrated in the star schema, the model strictly defines data types for all components: Surrogate Keys (PK/FK) and quantitative measures (qty, amount) are mapped as Integers (INT) and Decimals, while descriptive business attributes are stored as text strings (NVARCHAR).
Overall, this model allows the company to analyze sales performance across multiple dimensions, supporting better and more informed decision-making.

Data modelling methodology

To achieve the current dimensional design, the **Kimball 4-Step process** was followed. This approach ensures that existing business data is organized into a professional star schema optimized for high-performance reporting and analytical depth.

In the first step, the business process was identified as the core activity to be tracked. For this model, the process is Retail Sales, focusing specifically on the point-of-sale transaction. By selecting this process, the design ensures that the Data Warehouse provides the exact metrics required to analyze revenue and inventory movement across the organization.

The second step involved declaring the grain, which defines the level of detail represented by a single row in the fact table. The grain for this design is set at the individual product line item per transaction. This is the most atomic level of data possible. By choosing this detailed grain, the model allows for flexible reporting, ranging from total corporate sales down to the quantity of a specific SKU sold at a single register.

In the third step, the dimensions were identified to provide the "who, what, where, and when" for every sale. This involved utilizing existing tables for **Products, Stores, Locations,** and **Operators**, and developing a custom **Dim_Date** to complete the analytical set. These dimensions are primarily text-based and are designed to filter and group sales data. For instance, the date dimension supports "drilling down" from a year to a specific day, while the product dimension enables grouping by category or brand.

The final step was to identify the facts, which are the numeric measurements of the business process. For this model, quantity and amount were selected as the primary measures and stored within the central fact_sales table. Because these facts align perfectly with the defined grain, they are fully additive across all dimensions. This allows for seamless aggregation, such as calculating the total sales amount for a specific operator or a targeted geographic city.

The dimensional model design (star schema) puts the most important numbers in a central table and the descriptions in surrounding tables. This layout is built to make it very fast for a computer to find answers and very easy for a person to read a report. At the center is the **fact_sales** table, which records every item sold. It stores the "measures," which are the actual numbers we want to add up, such as the quantity of items bought. This table uses ID numbers, called foreign keys, to connect to all the other descriptive tables in the system.

The **Dim_Date** table is a special dimension that tracks exactly when sales happen. Instead of just showing a simple date, it is organized into a four-level **hierarchy**. This allows a manager to look at sales by the year, then break it down by quarter, then by month, and finally by day. This is helpful for seeing if the business is growing over time or if certain days of the week are busier than others.

The **Dim_Products** table describes what was sold and contains a three-level **hierarchy** that groups items together. Users can look at a broad category (like "electronics"), then a Subcategory (like "smartphones"), and finally the specific product Name. By organizing data this way, the company can see which types of products are making the most profit without getting lost in a list of thousands of individual items.

The **Dim_Location** table provides the geographic context for where the sales occur. It uses a three-level **hierarchy** that moves from the country down to the region and then to the specific city. This allows the business to see if one part of the country is performing better than another. This table works closely with the **Dim_Stores** table, which adds even more detail by describing the size of the shop and how much competition is nearby.

The **Dim_Operators** and **Dim_Point_Of_Supply** tables focus on the "who" and the "how" of the sale. The operators table includes information about the employee's experience and performance level, which helps the company see if more experienced staff sell more products. The point of supply table tracks the specific register or terminal used, which is helpful for fixing technical issues or seeing which machines are used the most often.

ETL Process

The ETL (Extract, Transform, Load) process is designed to automate the extraction, preparing and structure raw data for the analytical star schema. Using a shared workspace environment, the team can collaborate on handling data flows. To facilitate this, we began by creating our Microsoft Fabric workspace titled "**BI MAA 2025 Group 20"**. By systematically converting unprocessed source files into analyzable dimensions, the pipeline ensures data integrity and enables data-driven decision-making by providing actionable business insights.

Lakehouse

The initial step of the ETL process involves creating a lakehouse to serve as the designated repository for the project. This environment acts as the centralized destination for uploading all the necessary CSV files, required to analyze the Retail4All business process. By using the 'get data' feature, we uploaded every file individually, a method chosen to guarantee data integrity right from the source.

Warehouse

During the project, the Retail4all_DW_2026 data warehouse was built using the Kimball methodology to store sales data and support business decision-making. This approach was chosen because it offers high query performance. The system follows a star schema design, as previously mentioned, which makes searching for information faster and allows the warehouse to grow as the business collects more data.

To ensure a clean and well-working system, the development followed specific steps. First, existing tables were dropped to remove old versions of data. The architecture uses surrogate keys as unique identifiers and clear naming conventions to keep all information organized. To ensure data integrity, the central fact table (fact_sales) uses foreign keys to link to the various dimension tables, ensuring all tables communicate correctly.

Finally, we defined the granularity at the level of individual sales transactions, including the quantity and amount sold, providing a detailed and scalable framework for advanced reporting and high-performance queries.

Dataflows

To proceed with our ETL process, the next step involved the creation of dataflows. Each dataflow contains the data stored in the corresponding table from the lakehouse. Some transformations such as data type modifications and data merges were applied according to the business requirements for each table, supporting a better understanding of the data and further analysis. These transformations will be explained in more detail in the "data engineering" section of the report.

After creating the dataflows, they were assigned to a data destination, in this case the warehouse, where they were mapped to the corresponding tables, column by column, in order to load the data. This way, the raw data remains stored in the lakehouse, while the warehouse holds new clean and structured data.

At the end of this stage, we had a total of six dataflows, one for each dimension and one for the sales fact table.

Data Engineering

After establishing the initial dataflows, some targeted transformations were applied to shape the data into our final dimension and fact tables. Some of these steps are done in every table, such as putting the first row to headers and renaming columns to the names according to the star schema. A summary of the most important changes in some tables can be found below.

- DF_load_Dim_Stores : To consolidate the store information, the two separate CSV files (stores A and stores B) were appended together into a single dataset. During data cleaning, we standardized the online presence indicators by replacing ‘s’ with ‘y’. Additionally, any missing values were replaced with ‘u’ to fill all the lines and complete the dataset.

- DF_load_Dim_Location : To handle missing geographic data, we created a custom query acting as a district dictionary, which successfully mapped and replaced the missing district values.
- DF_Load_Dim_Date : This dimension was built by creating a blank query with a script that lists a continuous data range. This list was converted into a table starting with a ‘Proper Date’ column. From this base column, we extracted multiple hierarchical attributes: Day, Year, Month, Month_name, prefix_month, Quarter, Day_of_week, Day_name and Prefix_day. We applied localization adjustments to the Date column, changing the local settings to portuguese to ensure the dates followed the correct form (for example, in the column “Day_of_week” we replace the number 0 to 7, that means Sunday).

Columns based on ‘Proper Date’	Data Type
‘Day’	Whole number
‘Year’	Whole number
‘Month’	Whole number
‘Month_name’	Text
‘prefix_month’	Text
‘Quarter’	Whole number
‘Day_of_week’	Whole number
‘Day_name’	Text
‘Prefix_day’	Text

Table 2: Columns created based on ‘Proper_Date’.

- DF_Load_Fact_Sales : We mapped and validated all the foreign keys to guarantee they correctly align with their respective dimension tables. And finally, the pipeline was fully configured and is currently running successfully to load the data model.

Pipeline

The final step is to implement the ETL process to load the pipeline. This process involves a series of sequential steps to ensure the integrity and completeness of our data warehouse. Each phase must be successfully completed before proceeding to the next, so it's necessary to have intermittent pauses in the process as we wait for the completion of each stage. These waits allow us to verify the accuracy and completeness of each stage before advancing the subsequent ones, thereby safeguarding the reliability of our data warehouse.

For this, before loading the new data, we ensure that the data warehouse is empty of any existing data with the creation of 2 scripts where we are deleting the existing dimensions and the fact table. This is done to prevent overwriting or multiplying data when the pipeline runs.

Following the data warehouse cleanup, the next phase involves the dataflows for the existing dimensions of tables, in accordance with the transformations defined previously in each dataflow.

Lastly, with the dimensions tables created, we proceed to load the data into the fact table. Unlike dimension tables, the fact table contains transactional data and requires information from the dimension tables. Therefore, we prioritize loading the fact table last in the pipeline to ensure that all necessary dimension data is available for merging.

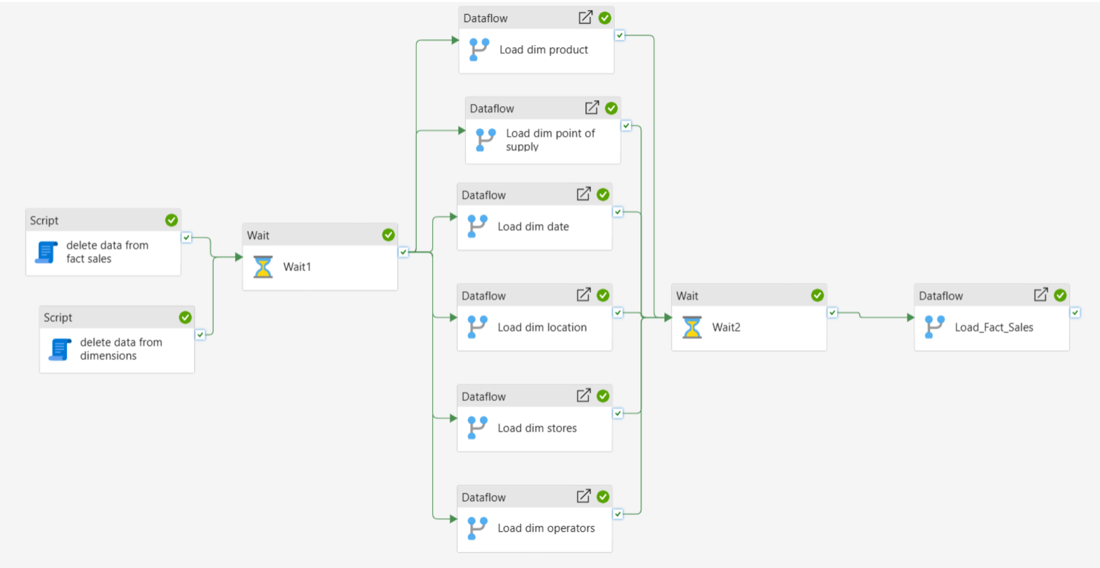

Figure 2: Pipeline design of the loading of the dim and fact tables.

Clustering Notebook

The clustering component of this project was developed to transform the data into actionable segments. In this case, the segments were defined at the store level. The process began with the ingestion of source data, where we imported two separate datasets (stores A and stores B) and performed a concatenation to unify all store records into a single dataset.

To ensure high data quality, we performed a brief analysis to confirm the absence of missing values, verify that all data types were correctly assigned, and ensure no duplicate records were present.

To prepare the data for the clustering algorithm, we separated the features into categorical and numerical variables, removing non-importing features like store ID. Then we applied one-hot encoding to convert categorical variables into a format compatible numerical format. Simultaneously, we applied scaling to normalize the range of numerical variables, preventing features with larger scales from dominating the model results.

With the features processed, we performed the optimal k analysis with the Elbow curve (Figure 3). Then we applied the K-Means clustering algorithm. This identified distinct patterns among the stores based on their attributes that were easy to analyze when we plot the store segmentation using PCA (Figure 4). The resulting labels were joined to the original store ids as a new column named cluster.

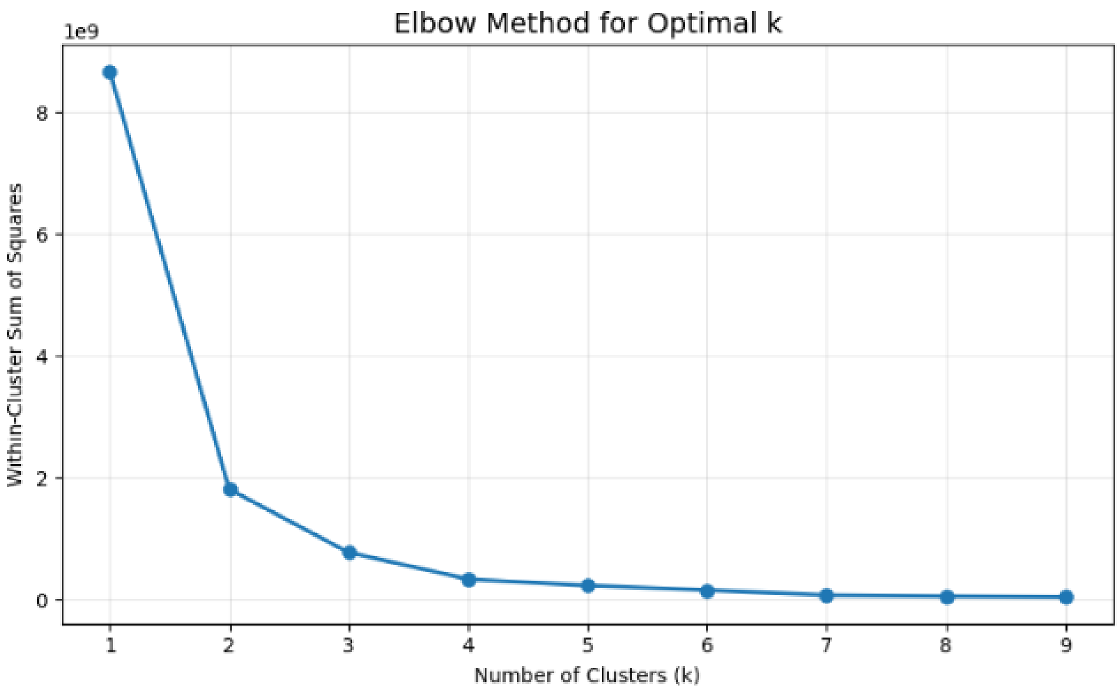

Figure 3: Elbow method for the Optimal k.

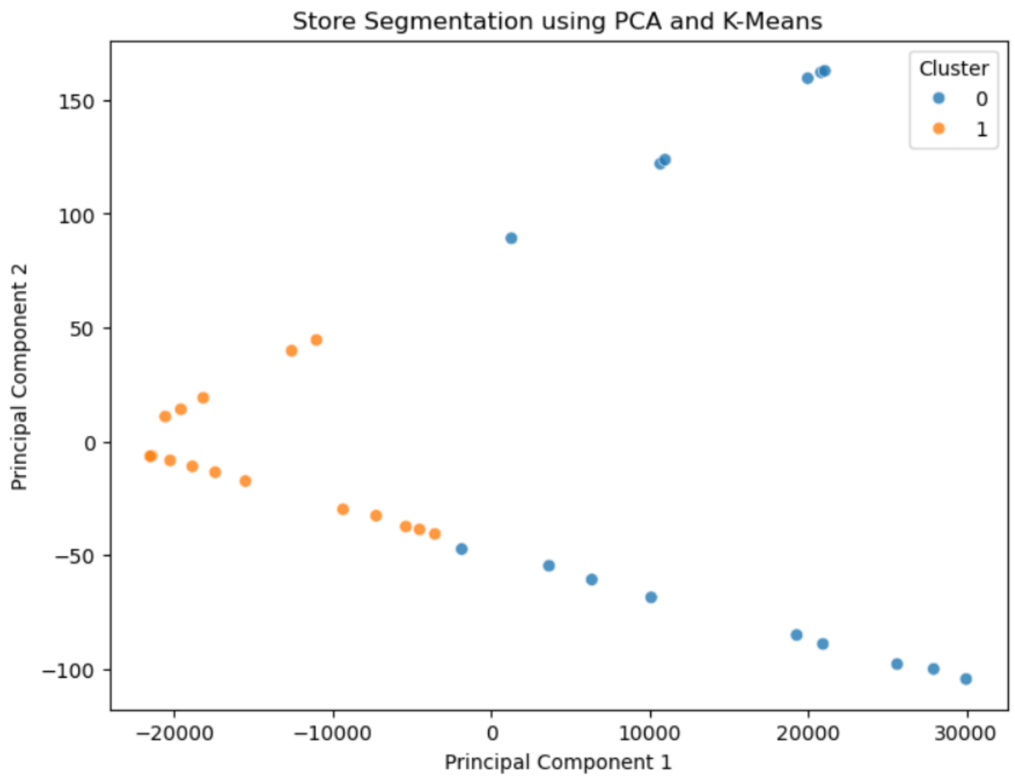

Figure 4: Store Segmentation using PCA and K-Means.

The results were saved at the end of the notebook directly to the lakehouse. Subsequently, the new table, containing the store ids and their associated clusters, was saved into the warehouse.

Finally, we integrated this entire workflow into the project pipeline. First, integrate script to delete any existing data in the dimensions table, then add the notebook activity to run the notebook created. Then finally add the dataflow for the new dimension created in the notebook, in accordance with the transformations defined previously in each dataflow.

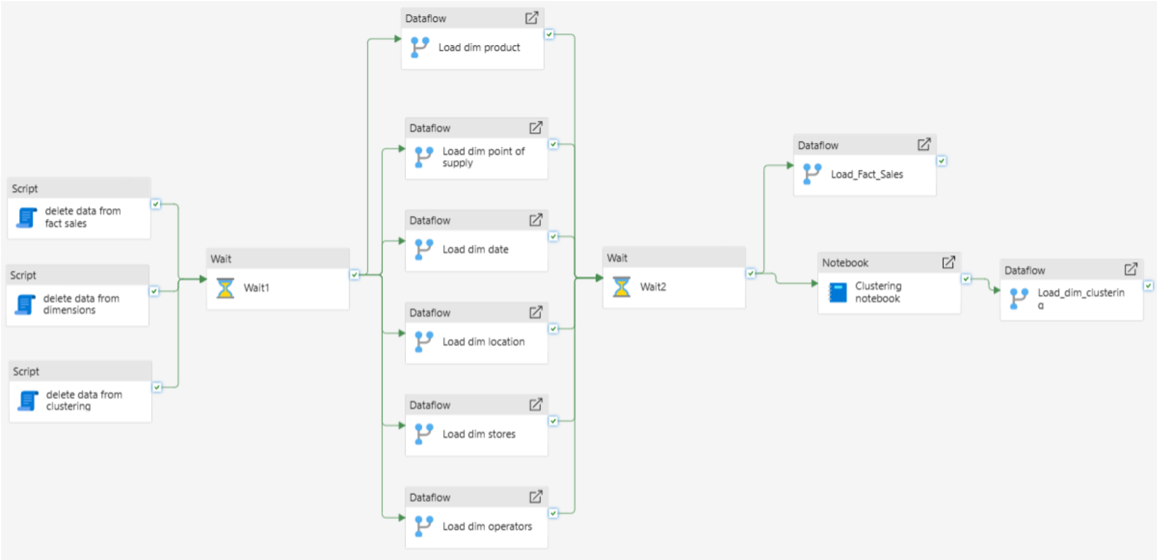

Figure 5: Pipeline design after the notebook phase.

Incremental load

After the initial full load of all the tables, an incremental load was implemented for the Sales fact table. This decision was made because the sales table is constantly expanding as new transactions are registered, and re-processing all the data on every load would become computationally inefficient over time. By implementing the incremental load, only the new or recently modified records are loaded into the warehouse, while relevant historical data remains untouched.

To set up this process, a new dataflow was created specifically for this purpose and some adjustments were made in the warehouse.

In the warehouse, the table structure was aligned with the output of the new dataflow, and the reference date column (‘Proper_Date’) was correctly defined. This allows new records to be added without conflicting with existing ones.

On the dataflow side, the new dataflow named ‘DF_Load_Incremental’ has an output that mirrors the original fact table, but includes the ‘Proper_Date’ column, from the date dimension, which is used as the main time reference.

Then, a ‘RangeStart’ and ‘RangeEnd’ were defined to determine the time window of the data being processed. These values were chosen based on the time of coverage of the problem. Next, a filter was applied so that only new records within these ranges were loaded.

Finally, the load configurations were defined. "Extract data from the past" was set to seven years, which establishes the maximum time window eligible for loading and anything older is automatically discarded. The "Bucket size" was set to "Quarter", dividing this window into 28 quarterly partitions. During each load, the system only reprocesses buckets containing new or modified records, which makes the process more efficient.

As with the other dataflows, the warehouse was defined as the data destination for ‘DF_Load_Incremental’.

The last step of the incremental load was to integrate it into the pipeline. This was created in a sequential way, so the best way to load the new dataflow was to add it after all the dimension tables are loaded. This ordering ensures that the full dimension tables are completely loaded before the incremental refresh runs, guaranteeing data consistency and avoiding conflicts during execution. Since the native Incremental Refresh feature of Fabric automatically manages the load control internally, no additional Lookup activities or control tables were required in the pipeline.

Additionally, two Email notification activities were integrated into the pipeline to monitor its execution status. If the load process is successfully completed, a ‘Notify_success’ activity automatically sends an email confirming that all data was loaded correctly. In case of failure, a ‘Notify_failure’ activity is triggered, sending an alert email to notify that an error occurred during the pipeline execution, allowing for a swift investigation and resolution. This proactive monitoring mechanism ensures that the pipeline status is always tracked, particularly useful when scheduling automatic loads.

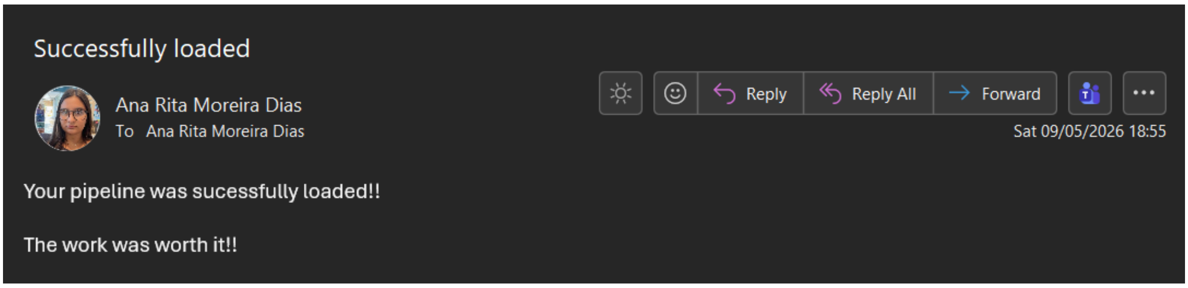

Figure 6: Email of successful pipeline running.

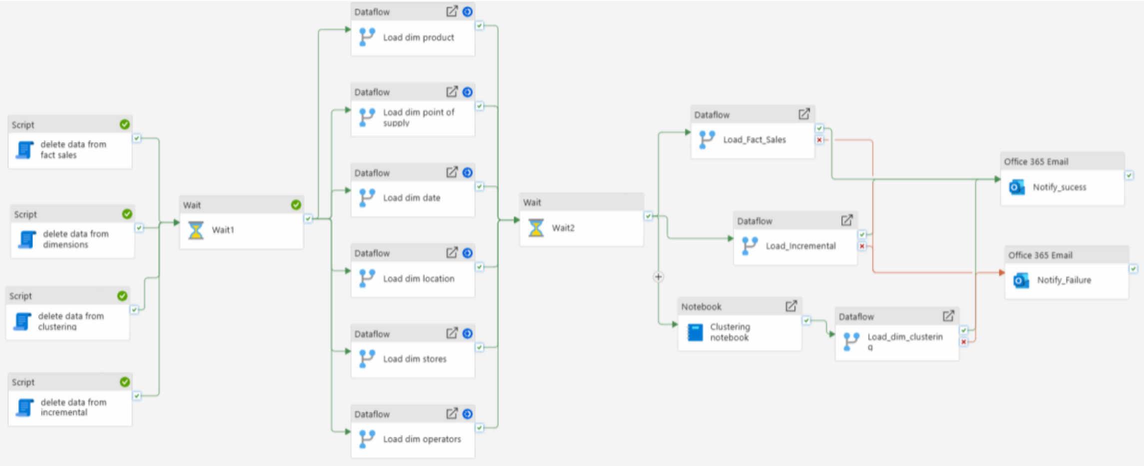

Figure 7: Final pipeline design.

Semantical Model

Connection to Power BI Service

To transform the structured data residing in the data warehouse into a powerful and interactive analytical tool, establishing an efficient connection for reporting was a critical step. Unlike traditional methodologies that rely on exporting data to a local environment, this project leveraged the native cloud capabilities of Microsoft Fabric. Several connectivity approaches were evaluated to ensure alignment with Retail4All's specific business requirements and the analytical complexity of the model.

1. Traditional Import Mode and Direct Query (Evaluated)

Traditional Import Mode ingests data directly into Power BI's in-memory engine, which provides fast query performance but creates data duplication and requires local processing power. Direct Query, on the other hand, queries the source at runtime but severely restricts the use of complex DAX functions, particularly advanced Time Intelligence calculations and dense ranking formulas. Since our model heavily relies on complex measures (for example, Revenue YoY %, and Operator Rank by Year), a standard Direct Query approach was not ideal, and downloading data locally via Import Mode countered our centralized cloud-first architecture.

2. Native Semantic Model Connection (Chosen Approach)

To maximize our analytical capabilities and maintain streamlined architecture, we opted to build the report directly in the Power BI Service, connecting seamlessly to the native Semantic Model generated by our Microsoft Fabric data warehouse. This strategic cloud-native choice provided several key advantages:

No movement of data and high performance: By authoring the report directly in the cloud via the browser, we bypassed the need to download massive datasets to a local client. The semantic model leverages Fabric's robust backend engine, ensuring that our complex DAX measures are computed instantly without local hardware bottlenecks.

Seamless pipeline integration: Because the report is directly linked to the workspace semantic model, any updates triggered by our automated ETL pipeline and incremental load strategy are immediately reflected in the dashboard. This eliminates the need for a secondary semantic model to refresh schedules, maintaining a single, real-time source of truth.

Cloud-native collaboration and security: Building the report directly in the shared workspace environment facilitates real-time team collaboration.

As an additional development (bonus), a custom report theme was defined in a JSON file created by the group. To apply it, Power BI Desktop was used to import the JSON theme into the report, ensuring a consistent and personalized color palette and visual identity across all pages. The same theme file is reused throughout the report, which guarantees that every section aligns with professional business intelligence standards and reinforces visual coherence. In addition, custom navigation buttons were configured to allow the user to move seamlessly between the report pages, as well as dedicated buttons on the **Product Performance** page that let the user switch the active measure between Total Revenue and Total Units Sold.

Model Optimization

Following the initial model configuration, a set of optimization procedures was implemented to enhance both the model's usability and the readability of the visualizations.

The process began with the definition of the dim_date as a date table, enabling the time intelligence functions to work correctly across the model. Additionally, the columns regarding the month and day_name were formatted to be sorted according to the month number and day of the week, which facilitates the readability of the visualizations by presenting the information in their natural order.

Subsequently, hierarchies were established across the relevant dimension tables. Starting with dim_date, the year hierarchy was created ranging from the higher level "Year" down to the smallest item "Day", following the sequence: Year → Quarter → Month Name → Day. Similarly, the dim_location table includes the hierarchy: Country → Region → District → City, representing the natural geographic aggregation. In the dim_product table, two hierarchies were defined, one more than what was initially planned. The first is the category hierarchy: Category → Subcategory → Product Name, and the second is the brand hierarchy, with only two levels: Brand → Product Name, as a single brand can encompass multiple products. Using both hierarchies supports different analytical perspectives: one focused on product classification and the other on the brand structure. Finally, the columns that are part of the hierarchies were hidden in their original non-aggregated form.

In fact sales table, the major optimization was the Amount format from number to currency that was defined as Euro. The same procedure was done for the sales price and margin in the products table.

Summarization rules were also applied to certain columns according to their most relevant aggregation form. For continuous numerical features such as ratings, rankings and average monthly sales, the chosen aggregation was “Average”, as these metrics are more meaningful when analyzed as averages across records. For categorical features, such as the online channel indicator and store size, “Count” was selected to enable frequency and distribution analysis across categories. For years in the company, “Sum” was selected as the default aggregation, as it provides an understanding of the teams' total accumulated experience.

For all the tables, the identification keys, including both surrogate and business keys, were hidden. Some additional fields that were discussed between the group not to be used in the further stage (Team, Discontinued and Postal_code) were also marked as hidden. However, it is important to reinforce that these columns are still part of the model and can be used at any time if the business analysis requires it.

Measures Creation

To support a comprehensive analysis of Retail4All's business performance, a set of DAX measures were created covering key areas such as revenue, product performance, operator efficiency, and geographic distribution. These measures were designed to answer the defined business questions and provide actionable insights across all report pages. The following DAX measures were created:

Name	Description	Type
Total Revenue	Total sum of revenue across all sales transactions.	KPI
Total Units Sold	Total sum of units sold across all transactions.	KPI
Brands	Number of distinct brands available in the product catalogue.	KPI
Categories	Number of distinct product categories.	KPI
Products	Number of distinct products in the catalogue.	KPI
Subcategories	Number of distinct product subcategories.	KPI
Online Sales	Total revenue generated exclusively by online stores.	KPI
Physical Sales	Total revenue generated exclusively by physical stores.	KPI
Number of Operators	Number of distinct operators with registered sales in the selected context.	KPI
Avg Margin	Average margin across all products in the catalogue.	KPI
Avg Revenue per Product	Average revenue generated per product, calculated as Total Revenue divided by the number of products.	KPI
Avg Years in Company	Average number of years operators have been working at the company.	KPI
Avg Customer Rating	Average customer satisfaction rating per operator (scale 1–5).	KPI

Avg Revenue per Operator	Average revenue generated per active operator in the selected context.	KPI
Avg Units per Operators	Average units sold per active operator in the selected context.	KPI
Avg Revenue per Store	Average revenue per store, calculated by iterating each store individually.	KPI
Avg Revenue per City	Average revenue per city in the selected context.	KPI
Avg Revenue per POS	Average revenue per point of supply in the selected context.	KPI
Avg Monthly Revenue	Average monthly revenue, calculated per year and month to avoid distortions across the years.	KPI
Avg Monthly Units	Average units sold per month, calculated per year and month.	KPI
Avg Monthly Revenue per Year	Average monthly revenue aggregated per year, useful for year-over-year comparison.	KPI
Avg Day of the Week Revenue	Average revenue per day of the week, calculated per year and day to identify weekly patterns.	KPI
Selected Operator Revenue	Revenue of the selected operator, ignoring all filters except the operator ID.	KPI
Districts with Sales	Number of distinct districts that have registered sales in the selected context, reflecting the breadth of the active geographic network.	KPI
Revenue PY	Revenue of the previous year relative to the selected one, dynamically calculated using VAR to ensure the correct filter context.	Time Intelligence
Revenue YoY %	Percentage change in revenue compared to the previous year where positive indicates growth and negative indicates decline.	Time Intelligence

Revenue Target YoY	Revenue for the same period in the previous year, used as the target value in the KPI visual for annual comparison.	Time Intelligence
Operator Rank by Year	Dense ranking of operators by total revenue in the selected year, ordered from highest to lowest.	Complex
Top Store Revenue Contribution	Captures the performance disparity between the best and worst performing stores, expressed as a share of total revenue where the higher the value, the greater the inequality between stores.	Complex
Top District Contribution	Captures the share of total revenue concentrated in the best-performing district, expressed as a percentage where the higher the value, the greater the geographic concentration of revenue.	Complex

Table 3: Dax measures.

Final Result:

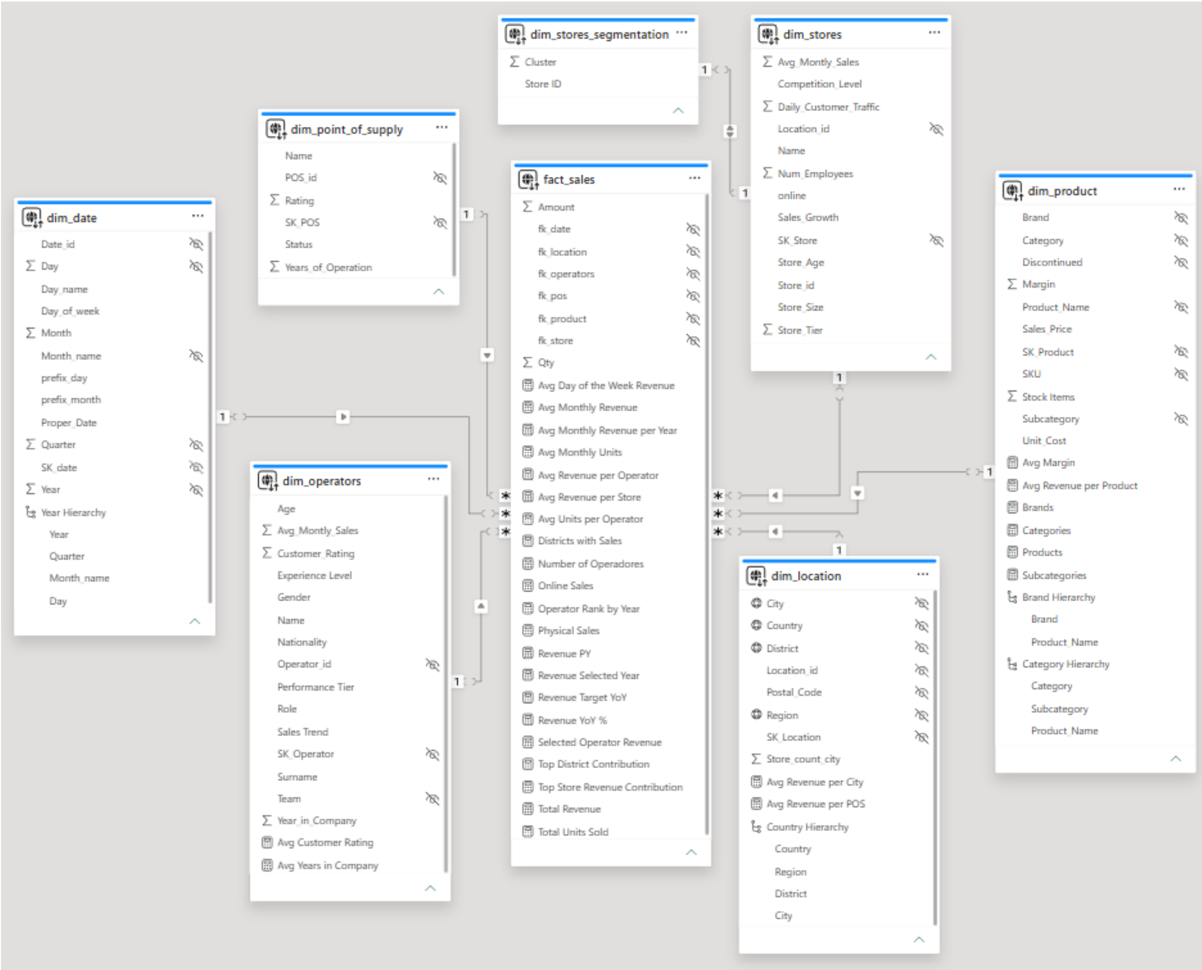

Figure 8: Final Semantical Model.

Power BI Report

Taking into account the business needs and questions defined in the first report, the Power BI solution was structured into thematic pages, each with its own narrative, so that the principle of proximity is respected, and a cohesive storytelling is maintained throughout the report. The navigation bar on the left allows the user to move between the different sections: **Executive Overview**, **Product Performance**, **Operator Performance**, **Channels and Supply Points**, and **Location and Business Drivers**.

The **Executive Overview** functions as the central hub of the project, presenting a high-level summary of key business metrics that provides an immediate, intuitive view of the company's current performance. Throughout the development process, we prioritized visual coherence by applying a personalized color palette across all report sections, defined in a custom JSON file, ensuring that all report sections remain consistent and align with professional business intelligence standards.

To address the specific business questions, we have developed the **Product Performance** section, which allows for a detailed investigation into sales trends by category, subcategory, and individual item, while the **Operator Performance** section shifts the focus to the "who" of the business, evaluating employee contributions, performance tiers, and experience levels.

To optimize operations, the **Channels and Supply Points** section provides visibility into the quality and longevity of supply channels, while the **Location and Business Drivers** section offers the tools necessary to perform regional analysis and examine store-level performance indicators, such as traffic, store size, and competition. Throughout the report, we have implemented essential analytical features to empower Retail4All's decision-making. Users can perform multidimensional time-intelligence analysis, ranging from daily transactions to year-over-year comparison, to track growth and identify performance fluctuations.

Furthermore, we have integrated the results of our advanced analytics, allowing for a strategic view of store performance profiles. This hierarchical structure ensures that users can seamlessly navigate from high-level corporate insights down to the granular detail of sales transactions, fulfilling the requirement for both strategic and operational visibility.

Cover Page

Figure 9: Cover Page Layout.

The report opens with a cover page that works as the main entry point and menu of the solution. Besides presenting the project's title and visual identity, it contains a page navigator that lists all the report sections (Executive Overview, Product Performance, Operator Performance, Channels and Supply Points, and Location and Business Drivers) and lets the user jump directly to any of them with a single click. This same navigation logic is then carried into every page through the navigation bar on the left, so the user can move freely between sections at any point, keeping the whole report cohesive and easy to browse.

Executive Overview

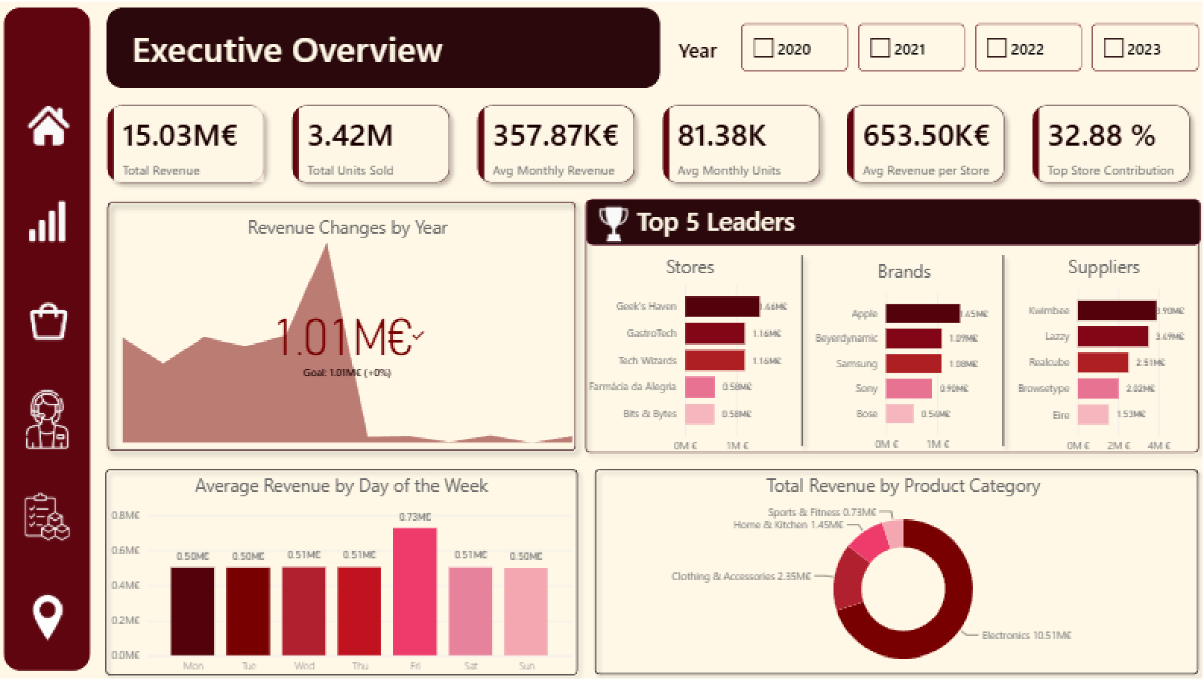

Figure 10: Executive Overview Layout.

The **Executive Overview** is the entry point of the report, and it works as an executive summary of the entire project. Its main objective is to provide a quick, intuitive and high-level view of the company's overall performance, designed for decision-makers who typically have limited time and need a direct, consolidated picture of the business before drilling down into more detailed sections.

This page contains a year slicer at the top, which lets the user select one or more years (2020, 2021, 2022 and 2023). This slicer controls the entire page, so every visual recalculates according to the selected period and enables year-over-year comparison directly from the summary view.

Immediately below, there is a row of six KPI cards which present the headline metrics of the business: Total Revenue, Total Units Sold, Average Monthly Revenue, Average Monthly Units, Average Revenue per Store and Top Store Contribution. These cards give an instant read on the size and concentration of the business. Average Monthly Revenue and Average Monthly Units were computed per year and per month to avoid distortions when several years are selected at the same time, ensuring the value shown is a true monthly average rather than a simple division of the total by the number of rows. The Top Store Contribution is expressed as a percentage and captures the share of total revenue concentrated in the best-performing store, acting as an inequality/concentration indicator rather than a simple total.

Below, on the lower areas of the page are organized into four analytical visuals. First a Revenue Changes by Year (KPI visual), that combines the year-over-year trend with the actual value against a target. This visual uses a native KPI with three elements: the indicator, which is the Total Revenue, the trend line, driven by the month prefix, so the trend is read chronologically, and a target represented by the metric Revenue Target YoY. The target is the revenue of the same period in the previous year, which turns the visual into an immediate year-over-year reading: the percentage shown next to the value tells the user instantly whether the business is growing or declining versus the comparable period.

Next to the KPI visual we have a panel that shows the Top 5 Leaders, grouping three horizontal bar charts side by side: Top 5 Stores, Top 5 Brands and Top 5 Suppliers all ranked by Total Revenue. Horizontal bars were chosen because the category labels (store, brand and supplier names) are text-heavy and read more naturally on the vertical axis, and a Top N filter was applied to each chart to keep the focus on the five highest contributors. Because all three charts respect the Year slicer, the ranking dynamically updates.

Below the KPI visual there is an Average Revenue by Day of the Week in a column chart, to expose weekly demand patterns. This visual sort the day by Monday-to-Sunday, so the week's read is in natural order rather than alphabetically. The measure analyzed here is calculated per year and per day, which means it represents an average and not an accumulated total, allowing a fair comparison of which weekdays are systematically stronger.

Finally, below the Top 5 Leaders panel, there is a Total Revenue by Product Category in a donut chart, showing the contribution of each category in the total revenue (Electronics, Clothing & Accessories, Home & Kitchen, Sports & Fitness).

All visuals on the page are cross filtered, which means that clicking on a category in the donut, a day in the column chart, or an entity in any of the Top 5 charts filters the rest of the page, allowing the executive to interrogate the summary interactively.

This page is the one that most directly answers the high-level business requirements, giving the company important insights. If the user considers the full period of data (2020-2023), he can see that the company recorded approximately 15.03M€ in Total Revenue and 3.42M units sold, with a monthly average of 357.87K€ of revenue and 81.38K of units. The Average Revenue per Store stood at 653.50K€ and the Top Store Contribution of 32.88% shows that roughly a third of all revenue is concentrated in a single store, which is a clear signal of dependency that management should monitor.

When the user chooses the year slicer to the period of 2020-2021, the same KPIs read 7.34M€ in Total Revenue, 1.77M units sold, 305.92K€ average monthly revenue and a Top Store Contribution of 28.63%, which demonstrates both the growth of the business over time and the increasing concentration of revenue in the leading store.

This page contributes to answering our business questions, where the Total Revenue by Product Category donut, combined with the Year slicer and its drill-down to subcategory and product, answers the question of total revenue and units sold by product category across time periods at the summary level. Electronics is consistently the dominant category (10.51M€ over the full period, versus 2.35M€ for Clothing & Accessories, 1.45M€ for Home & Kitchen and 0.73M€ for Sports & Fitness), while the Total Units Sold KPI complements the revenue view. The more granular breakdown is then explored in the Product Performance page.

For the question which operators generate the highest sales, although the dedicated answer lives in the **Operator Performance** page, this pages frames operator-related KPIs that set the context for that analysis.

To answer which Points of Supply generate the majority of the sales, the Top 5 Suppliers bar chart answers this directly, ranking the leading points of supply by revenue (over the full period, Kwimbee at 3.90M€ and Lazzy at 3.49M€ lead the channel ranking), making it immediately clear which channels drive the bulk of the business.

Finally, for which locations/stores drive the highest revenue the bar chart of the Top 5 Stores, the Average Revenue per Store and Top Store Contribution KPIs answer part of the question, identifying the leading stores (for example, Geek's Haven at 1.46M€ over the full period) and quantifying how concentrated revenue is among them.

In short, the **Executive Overview** consolidates the headline numbers and the leading dimensions (category, store, supplier, weekday, year) on a single screen, so that a decision-maker can see the state of the business in seconds and then navigate to the specialized pages for deeper analysis.

The main takeaways from this page are that Retail4All has grown consistently across the analyzed years, that Electronics is by far the strongest product category, which makes sense, once they are normally the more expensive products. Also, we takeaway that revenue is significantly concentrated in a small number of stores and supply channels, and that average revenue per day of the week is fairly stable, with identifiable peak days. These insights, all reachable in seconds and adjustable through a single Year slicer, give Retail4All's administration exactly what was requested in the first page: a faster, data-driven way to monitor performance and to decide where to focus a more detailed investigation in the remaining sections of the report.

Product Performance

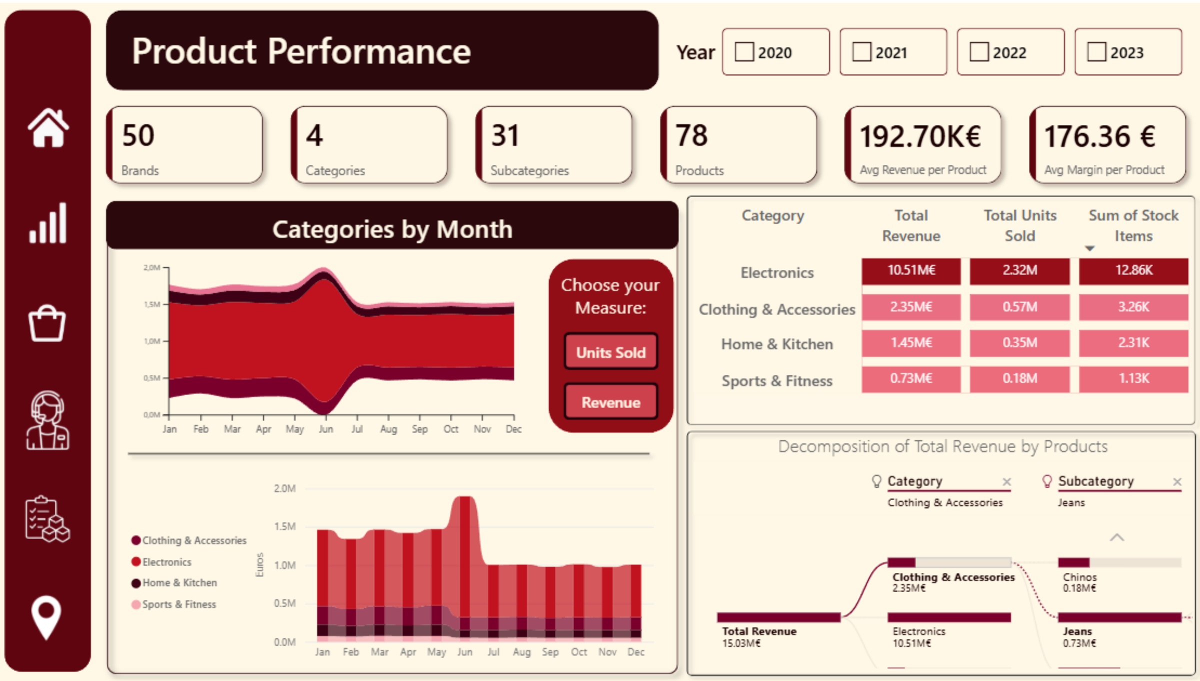

Figure 11: Product Performance Layout.

The **Product Performance** page was designed to answer, in depth, the questions related to the product dimension of the business, namely the analysis of revenue and units sold by product category, subcategory and individual product across different time periods. While the **Executive Overview** gives a high-level snapshot of the category mix, this page allows the user to drill into the catalogue and understand exactly which categories, subcategories, brands and products drive the company's performance and how that performance evolves over time.

This page keeps the same visual identity, navigation bar and overall structure as the rest of the report, so the user experiences a consistent layout when moving between sections. At the top, there is the same Year slicer (2020, 2021, 2022 and 2023) that controls the whole page and enables year-over-year comparison. For some graphs, there is a Choose your Measure control that lets the user switch the active metric between Total Revenue and Total Units Sold, so the same visuals can be read either in monetary or in volume terms. This was the central design decision of the page: by making the measure itself dynamic, the user can flip some parts of the page between value and volume without needing two separate sets of visuals, which keeps the page compact and lets the analyst compare the two stories on exactly the same layout. This distinction matters, since a category can be a top seller by units but not by revenue, and vice versa.

Immediately below the year slicer, there is a row of KPI cards that summarizes the structure of the catalogue, showing the number of Brands, Categories, Subcategories and Products, together with the Average Revenue and the Average Margin per Product. The Average Revenue per Product is calculated as Total Revenue divided by the number of distinct products, and the Average Margin across the catalogue. Both react to the Year slicer giving immediate context to whatever subset of products is being analyzed.

The central and lower areas of the page are organized into several analytical visuals. First, there is a Categories by Month stream graph, which was chosen to show the composition and the rhythm of the categories at the same time. Each band represents a category. The horizontal axis is driven by the month in chronological order, and the height of each band reflects the selected measure. The flowing shape makes seasonal patterns and the relative weight of each category very easy to perceive, while still respecting the Year slicer for period-specific analysis. Next to it below, a ribbon chart by month and category complements the stream graph by emphasizing the ranking over time rather than the absolute composition. For each month it stacks the categories and connects them with ribbons, so the user can immediately see whether a category gains or loses position from one month to the next. It uses the same month's axis and the same dynamic measure to stay consistent with the stream graph, but the analytical question it answers is different: not how big each category is, but whether each category is climbing or falling in the ranking. These two graphs are the ones controlled by choose your measure button.

The page also includes a decomposition tree, which is an interactive visual for this section. It analyses the selected measure and lets the user explain it by Category, then Subcategory, then Product, expanding the branch they are interested in. This directly supports the granular part of the business question, moving from a broad category down to the specific products that drive it, and it does so in an exploratory, self-service way, so the user is not limited to a pre-defined drill path.

Finally, there is a matrix that crosses the product hierarchy on the rows and shows Total Revenue, Total Units Sold and Stock Items together. This is the visual that most precisely answers the requirement to read total revenue and units sold by product category.

As in the **Executive Overview**, all visuals on the page are cross-filtered, which means that selecting a category in the stream graph, a node in the decomposition tree, or a row in the matrix filters the rest of the page, allowing the analyst to interrogate the catalogue interactively.

This page is the one that most directly and completely answers the business question of what the total revenue and units sold are by product category across different time periods, and it extends it down to subcategory and individual product. The combination of the matrix and the dynamic measure mean the user can read, for any selected year or combination of years, the exact Total Revenue and Total Units Sold for each category. Consistent with the **Executive Overview**, Electronics emerges as the dominant category in revenue terms across the analyzed period, while the dynamic measure switch lets the analyst check whether that dominance also holds in units sold or whether other categories punch above their weight in volume.

The stream graph and the ribbon chart together explain the temporal behavior of the categories, where the stream graph reveals the seasonal shape and the relative weight of each category month by month, and the ribbon chart shows whether any category systematically climbs or drops in the ranking over the year. This is valuable for planning, because a category that is stable in revenue but rising in units (or vice versa) signals a change in pricing or demand mix that management should react to.

The decomposition tree then turns the high-level category figure into an explanation, since starting from Total Revenue the user can expand into the subcategories and the specific products that account for the largest share, identifying the catalogue's true revenue.

The Average Revenue per Product and Average Margin KPIs put this into a profitability context, and the Stock Items column in the matrix links demand to inventory, helping to flag products that sell well but may be under-stocked, or products that hold stock without generating proportional revenue.

In short, the Product Performance page answers the product-related business question not as a single static number but as a fully explorable structure: by category and over time through the stream graph, the ribbon chart and the matrix, by hierarchy depth through the decomposition tree, and by value versus volume through the dynamic measure, all controllable through the Year slicer.

The main takeaways from this page are that Electronics is the clear leader in revenue, that the relative position of the categories can differ noticeably depending on whether revenue or units are considered, that category performance follows identifiable monthly patterns, and that the decomposition tree and the matrix make it straightforward to trace any aggregate figure down to the specific subcategories and products responsible for it. Together with the Stock Items and margin information, this gives Retail4All a precise, self-service tool to monitor which products drive the business and to support pricing, assortment and inventory decisions, fully answering the product-related question raised in the first report.

Operator Performance

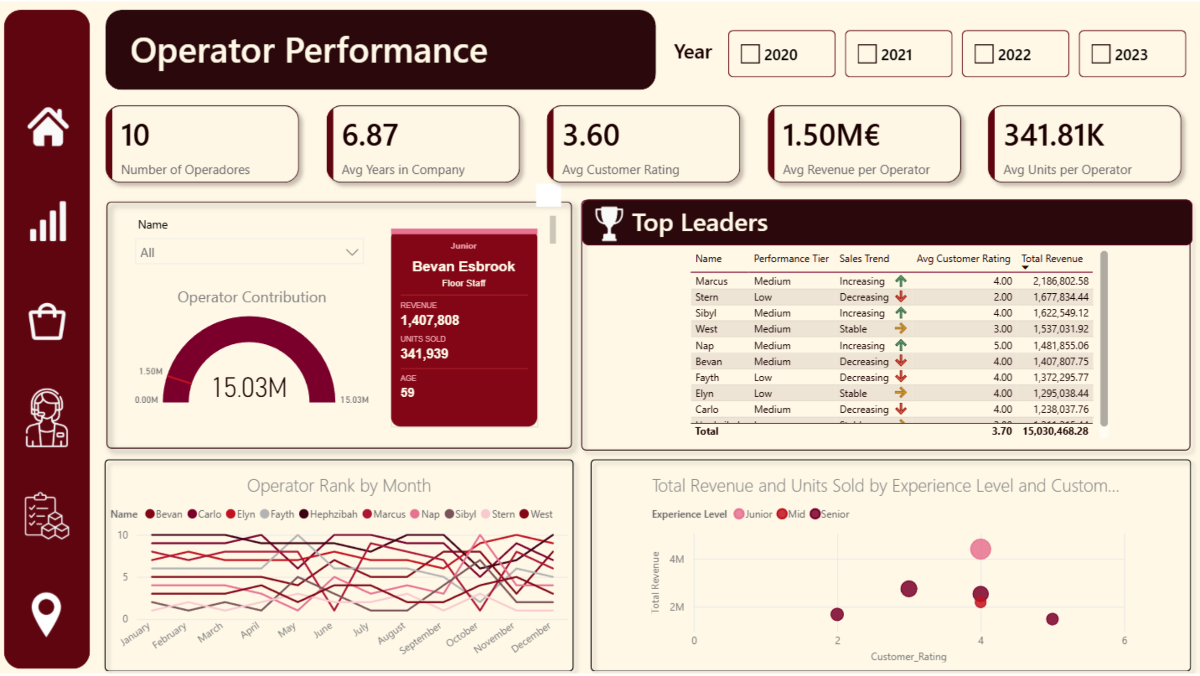

Figure 12: Operators Page Layout.

The **Operators Performance** dashboard aims to provide a general overview of the operators' performance during different time periods. Similarly to the other dashboards, this dashboard also includes an interactive panel with the years that updates the information according to the selected time period, as well as some business cards with the main highlights such as the total number of active operators, meaning the ones that have sales data/transactions associated, the average seniority of the operators, the average customer rating, the average revenue per operator and the average number of units sold by operator each year. The dashboard is split into 4 major sections.

The first section provides a detailed analysis of individual operators. Using the slicer option, the manager is able to select the desired operator and check their business card, which displays their profile information such as their level in the company (Junior, Mid or Senior), job title, age, revenue generated and number of units sold. This card was built using Deneb, a custom visual that must be imported into the report, which enables fully customized visualizations through the Vega-Lite specification language. In this case, the card layout was coded by defining layered text and rectangle marks, with each field from the data model mapped to a specific visual element within the card. Additionally, it displays a gauge measure that compares the selected operator's revenue with the team average, providing a clear visual indication of the operator's contribution to the company's overall results. The gauge was configured with a minimum value of zero, a maximum value corresponding to the total revenue of all operators, and a target line set at the average revenue per operator, meaning that the needle's position relative to the target line immediately indicates whether the selected operator performs above or below the team average.

The second section contains a table that is ordered by operator ranking in terms of sales. This table serves as an indicator of which operators might deserve compensation or a performance review. The table also provides additional information about their current status, including their current performance tier and sales tendency, as well as their customer rating, which provides a subjective measure of quality of service to balance the evaluation.

Although the table already displays the current sales trend, this trend is not always consistent and may only reflect a specific moment that is not contextualized within a broader timeframe. For this reason, a bump chart was added using the line chart visual to present the evolution of each operator's ranking during the different months of the year. When filtering multiple years, this graph can also indicate broader seasonal trends that can help frame each operator's performance. It can also be compared with store or location performance to better evaluate the factors that might influence it.

Finally, the last section shows a scatter plot combining total revenue and customer rating. This is grouped by experience level, with the size of the points influenced by total units sold. This enables us to evaluate the direct influence of experience level on total revenue and customers' perception of the service. Surprisingly, the results show that senior performance is inconsistent and does not always result in a positive customer experience. This contrasts with junior performance, which generates more revenue and maintains a solid customer rating throughout the four-year period. These results raise questions about whether experience directly translates into better performance in this context or whether other factors, such as role, team or store assignment, might be more influential drivers of sales and customer satisfaction.

Regarding the business question of which employees generate the highest sales across different time periods, the dashboard clearly identifies Marcus and Stern as the consistently top performing operators over the analyzed period, with a combined accumulated revenue that significantly surpasses the remaining team. Stern led the ranking in 2020, 2021 and 2022, while Marcus assumed the leadership in 2023 with 1,09M€, the highest individual yearly revenue recorded across all operators and all years. Together, the 10 active operators generated a total revenue of 15,03M€ over the four-year period, with an average of 1,5M€ per operator.

Additionally, other findings point to relevant patterns beyond the direct answer to the business question. Specifically, the current sales trend indicates that 4 operators are improving, 4 are declining, and 2 remain stable, highlighting that almost half of the team requires attention or strategic intervention to prevent further revenue deterioration. With 10 active operators, an average seniority of 6.87 years and an average customer rating of 3.60 out of 5, the team demonstrates a relatively experienced workforce with moderate customer satisfaction levels. Finally, the scatter analysis by experience level challenges the assumption that seniority guarantees better results, since senior performance proved inconsistent and not always tied to a positive customer experience. This suggests that experience may not be the main driver of performance, giving Retail4All management a data-driven basis to take actions regarding its workforce.

Together, these metrics provide an overview not only of the operators' sales performance but also of the clients' perception of the service, enabling comprehensive evaluation of each operator's contribution to the business.

Channels and Supply Points

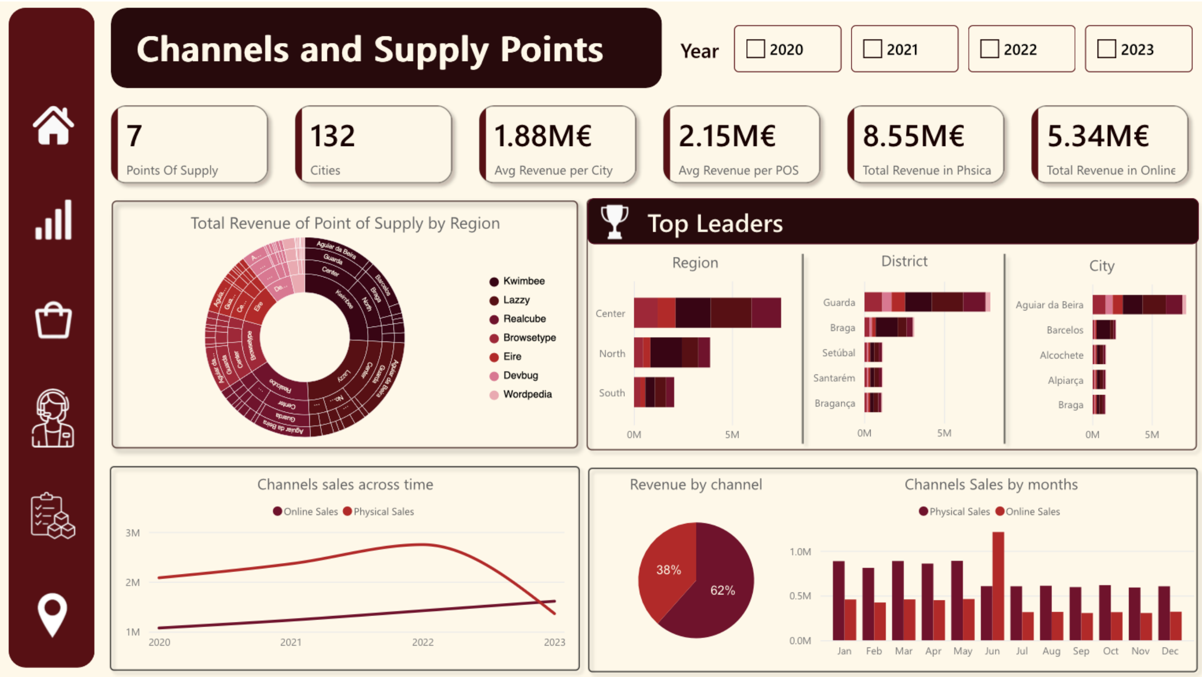

Figure 13: Channels and Supply Points Page Layout.

The **Channels and Supply Points** dashboard section is specifically engineered to address two core business questions: identifying the primary revenue-generating Points of Supply and evaluating the temporal performance differences between physical and online sales channels. Building upon previously established analytical measures, the dashboard leverages defined online and physical revenue DAX formulas to ensure precision and enable dynamic filtering within the Microsoft Fabric Power BI environment.

This page keeps the same visual identity and structure as the rest of the report, with a layout that prioritizes information density and intuitive navigation across both the suppliers' infrastructure and the sales performance. At the top, it integrates the same global Year filter (2020 to 2023) used on the other pages, so the whole page recalculates according to the selected period. Immediately below, a row of metric cards provides immediate context, showing the total number of supply points, the number of cities, the average revenue per city, the average revenue per point of supply, the total revenue in stores and the total revenue online. Then a Sunburst chart was utilized to map revenue distribution across regions and points of supply. This is complemented by the "top Leaders" section featuring three bar charts that rank performance across all 3 regions, the top 5 districts and the top 5 cities, using the points of supply as a legend to provide granular insight into network performance. These visuals indicate a highly concentrated revenue structure, where specific regional hubs significantly outperform others, allowing management to pinpoint the most critical infrastructure assets and focus strategic investments on the highest-performing supply points.

Following, to visualize how sales volume differs between channels over time, the bottom section of the page balances a line chart showing historical sales trends across time, a pie chart representing the total revenue split by channel, and a clustered bar chart that details seasonal sales by month and by channel. An analysis of these trends requires context regarding the dataset limitations, because the data spans from January 2020 through June 2023, so the monthly totals for the first half of the year include four years of data while the second half includes only three. Within this context, the observation that online sales outperform physical operations in June is particularly significant: despite the potential for volume bias in the first half of the year, this crossover highlights a distinct period where digital engagement demonstrably scales to exceed traditional physical retail performance.

Looking at the results over the full period (2020–2023), the page makes the structure of the supply network immediately clear. Retail4All operates through only 7 points of supply distributed across 132 cities, generating an average revenue per city of 1.88M€ and an average revenue per point of supply of 2.15M€. The Sunburst and the Top Leaders charts confirm the highly concentrated revenue structure: the Center region is by far the strongest, clearly outperforming North and South, while at a more granular level Guarda leads the district ranking and Aguiar da Beira the city ranking, with Kwimbee and Lazzy standing out as the dominant points of supply. This directly answers the first business question, pinpointing which channels and locations drive the bulk of the company's revenue.

On the channel side, physical stores generated 8.55M€ against 5.34M€ online, meaning physical retail still accounts for roughly 62% of total revenue versus 38% online. However, the line chart shows the online channel steadily closing the gap and overtaking physical sales in 2023, and the monthly view confirms a clear online peak in June, which reinforces the crossover described above and answers the second business question regarding how the balance between channels shifts over time.

In short, this page translates complex retail data into an actionable story, enabling the rapid identification of performance shifts and providing the necessary context to interpret seasonal anomalies against the constraints of the underlying dataset. The main takeaways are that revenue across the supply network is heavily concentrated in a few regional hubs and top-performing points of supply, which management can target for strategic investment, and that, although physical sales dominate overall, the online channel shows clear momentum, even overtaking physical retail in June, signaling the growing relevance of digital engagement for Retail4All.

Location and Business Drivers

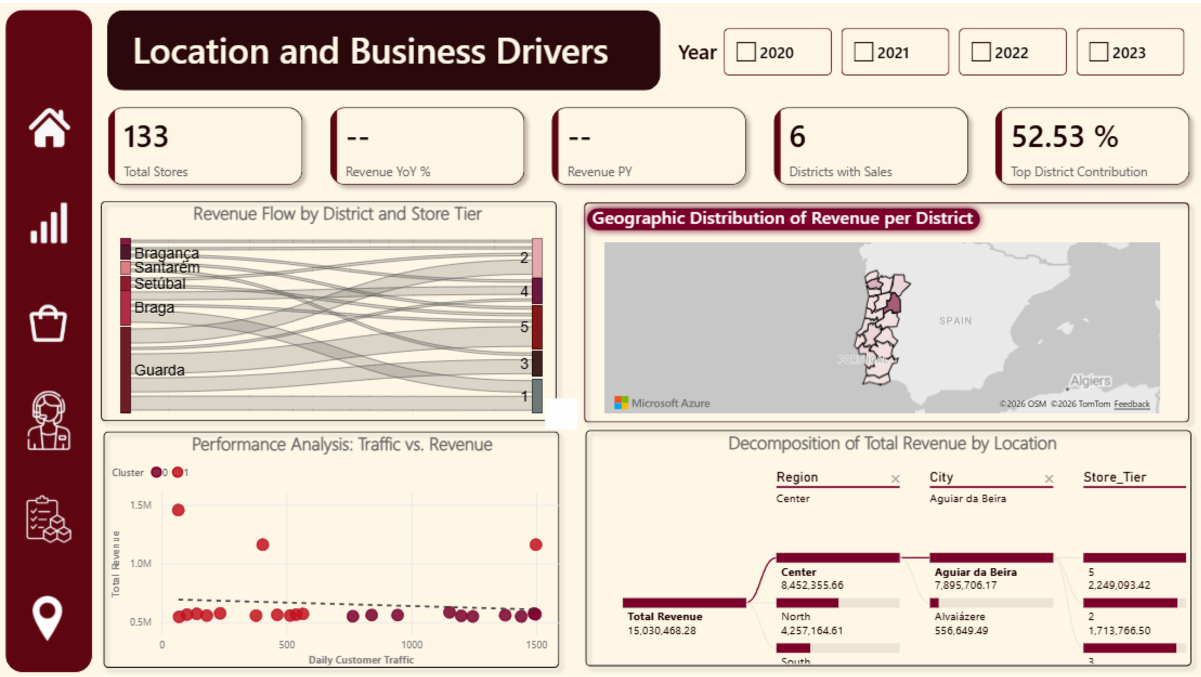

Figure 14: Location and Business Drivers Page Layout.

The **Location and Business Drivers** dashboard is designed to provide a deep dive into the geographical distribution of Retail4All's sales and the underlying factors that influence store performance. Specifically, this section was built to answer 2 core business questions: identifying which locations drive the highest revenue and growth and understanding how store performance varies when accounting for operational metrics, such as customer traffic, store size and competition level.

This page maintains the same visual identity and overall structure as the rest of the report, ensuring a consistent reporting experience. At the top, the Year slicer enables users to analyze regional trends over specific timeframes and perform year-over-year comparisons.

Directly below, a row of KPI cards establishes the baseline for the selected context: Total Stores, Revenue YoY %, Revenue PY, District with Sales and Total District Contribution.

To directly address the business question “Which locations drive the highest revenue and growth?”, the dashboard uses a combination of visual tools: the “Geographic Distribution” map, the “Decomposition of Total Revenue” tree and the “Revenue Flow by District and Store Tier” Sankey diagram.

The map was optimized with a color gradient (choropleth) to provide an immediate macro-level view of revenue density, while the decomposition tree acts as the primary exploratory tool, using artificial intelligence to dynamically break down the Total Revenue root into Regions, Cities and Store Tiers.

For instance, when filtering the data for 2022, the dashboard registers a Total Revenue of 4.53M€, representing a 16% growth (0.16 YoY) from the 3.91M€ revenue recorded in the previous year. The tree and the map clearly highlight the Center region as the leading geographic contributor. Remarkably, out of that 4.53M€ total, the district of Guarda and the city of Aguiar da Beira capture a significant concentration of sales, generating approximately 2.54M€ in revenue alone.

To bridge the gap between geography and store quality, the Sankey diagram was introduced to visually trace how revenue flows from specific districts (such as Guarda or Braga) into the different Store Tiers (1 through 5). This allows analysts to instantly verify if top-tier locations are effectively capturing the market share in key regions. The "Revenue YoY %" and "Revenue Previous Year" KPI cards complement this territorial analysis by instantly calculating the historical baseline and growth rate for any selected area.

To answer the final business question “How does store performance vary when accounting for store size, customer traffic and competition level?”, the "Performance Analysis: Traffic vs. Revenue" scatter plot serves as the primary analytical tool.

This visual plots Total Revenue against Daily Customer Traffic, illustrating the correlation between visitor numbers and financial success. The data points are color-coded by Cluster (made earlier in this project). As a key visual enhancement, a dynamic trendline was incorporated into this chart. This addition allows decision-makers to instantly benchmark individual stores against the expected revenue for their specific traffic volume, easily spotting overperforming or underperforming locations.

Furthermore, because the underlying data model was enriched with additional store attributes, the analyst can use the filter pane to cross-reference this scatter plot with "Store Size" and "Competition Level". This facilitates the isolation of variables and the detection of performance outliers. For example, the administration can quickly pinpoint highly efficient branches that generate substantial income despite being "Small" in size and located in high-competition areas, or recognize "Large", low-competition stores that underperform despite high daily customer traffic.

As with the rest of the report, all visuals on this page are fully cross-filtered. Selecting a high-performing cluster on the scatter plot immediately highlights where those retail units are located on the map, while choosing a specific city on the decomposition tree updates the scatter chart to display only the relevant local stores.

In summary, the Location and Business Drivers page successfully translates complex spatial and operational data into an actionable format. It provides Retail4All with the analytical framework required to evaluate whether a store's financial success is primarily determined by its geographic placement and lack of competition, or by genuine operational efficiency and high customer traffic.

Conclusion

In conclusion, this project successfully developed a business intelligence solution for Retail4All by transforming raw and fragmented transactional data into a structured and analytical dimensional model. This project provides Retail4All with the structural foundation necessary to navigate a highly competitive retail market with precision and clarity. By successfully centralizing three years of fragmented historical data through the Kimball methodology, we developed a robust star schema that integrates multiple data sources into a structured and organized format.

This architectural transformation was achieved by building a dedicated data warehouse supported by a lakehouse repository to act as the centralized destination for all source files.

To transform this raw data into actionable insights, we established a comprehensive ETL process using dataflows for every dimension and fact table. These dataflows allowed us to apply critical transformations, such as handling incorrect values, creating hierarchical attributes for time-intelligence, and merging datasets like stores A and B. This entire ecosystem is orchestrated through automated pipelines that ensure the integrity of the process by sequentially cleaning the warehouse, loading the dimensions, and finally updating the fact table and the incremental load. To ensure operational reliability, these pipelines include a final notification activity, sending automated emails to confirm whether the process has succeeded or failed.

Furthermore, the analytical value of this infrastructure is fully realized in the reporting layer, where a cloud-native semantic model was built directly in the Power BI Service to leverage high-performance DAX and complex time-intelligence calculations. This layer translates the structured warehouse data into an interactive, five-page Power BI report, comprising the Executive Overview, Product Performance, Operator Performance, Channels and Supply Points, and Location and Business Drivers sections. Through thematic storytelling and dynamic features like Year slicers, drill-down hierarchies, and measure-switching controls, this report provides a direct, intuitive, and self-service interface that specifically answers all defined business questions, from high-level corporate revenue trends down to granular store-level and product-specific performance metrics.

By enriching the model with advanced analytics like k-means clustering, the business can now move beyond simple observation to perform detailed store segmentation and performance tracking. Ultimately, this Business Intelligence solution transforms raw operational volume into a strategic asset, empowering Retail4All with the security, scalability, and technical infrastructure required to make faster, data-driven decisions that drive long-term growth.